

Phenotype Strands After SIMB

Expression, Morphology, and Metabolite Production: What Was Measured and What to Run Next

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Abstract

Six phenotype strands were run against one shared cell embedding: knockout expression, knockout expression jointly with a second knockout, morphology, betaxanthin production, free amino-acid pools, and beta-carotene production. Two of them were presented at SIMB 2026. This is the record of what each measured, against what ceiling, and with what scoring rule. The headline is that most of the strands are further from their ceilings than the manuscript's placeholder text assumes, that the expression strand contains no valid comparison between mechanisms because no run in it was ever trained to saturation, and that the metabolite strands converge and then overfit within a few hundred epochs, which is the opposite failure. One new measurement is reported: the free amino-acid profile of a deletion predicts that deletion's betaxanthin production at an out-of-fold Pearson r of 0.298, while tyrosine, the chromophore precursor, predicts it at 0.064. The closing section ranks what to run next and states what each option would settle.

How to read this. section 1 is the whole argument and every number a decision turns on. sections 2 to 6 are one section per strand, each ending in what it established and what it cannot say. section 7 collects the failures the strands share, which is where the leverage is. section 8 ranks the options.

Working notes: `notes/experiments.019-simb-multimodal.*.md`, `notes/experiments.020-cachera-betaxanthin.*.md`

Generating scripts: the `scripts/` directory of
`experiments/019-simb-multimodal/`, `experiments/020-cachera-betaxanthin/`, and
`experiments/023-metabolome-betaxanthin-joint/`

Leaderboard: `experiments/019-simb-multimodal/results/round_leaderboards.csv`

Section status:  ready  author-reviewed, keep stable  not done, agents may edit freely

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1 Summary ✗

1.1 Where each strand stands ✗

strand	ceiling	best	of ceiling	epochs (peak/last)	runs
Expression (Kemmeren, Sameith)	0.775	0.229	30%	4007/4105	163
Expression, masked-label objective	0.775	0.241	31%	9188/9997	16
Morphology (CalMorph, Ohya)	0.611	0.082	13%	27/65	23 (1 flat)
Expression and morphology, joint	–	0.051	–	28/70	32
Betaxanthin (Cachera)	0.914	0.372	41%	254/999	44
Betaxanthin with metabolome head	0.914	0.430	47%	176/496	46
Amino-acid pools (Mulleder)	–	0.211	–	402/999	31
Beta-carotene (Ozaydin)	0.544	0.132	24%	799/999	39

Table 1: Every strand, scored the same way. *best* is `roll_max`, the maximum of a centered five-point rolling mean of the validation curve, an upward-biased order statistic whose bias grows with epochs run, which is why the epoch column travels with it. *epochs (peak/last)* gives where the best run peaked and where it stopped: a peak in the last tenth means the run was cut off while still improving. Ceilings are measured by three different estimators because the three measurement types require them; the amino-acid cell is empty because Mulleder has one replicate per strain, and no ceiling is estimable. *flat* counts runs whose validation metric never left zero.

Read the epoch column with the score column. Expression and beta-carotene peak in the last tenth of their best run, so those numbers are lower bounds set by the compute budget. Morphology and the expression-morphology joint arm peak at epochs 27 and 28 of runs that stopped at 65 and 70, which is not a result of any kind. Betaxanthin and amino acid peak in the first half of a 999-epoch run and then overfit.

The two betaxanthin rows are not comparable to each other: the joint row trains on 3,832 strains against the plain row’s 3,698, because the joint configuration restricts to genotypes carrying both phenotypes. The within-grid paired comparison in table 13 is the valid form of that contrast, and it comes out negative.

1.2 Eight findings ✗

1. **The expression strand contains no valid comparison between mechanisms.** All 67 scored arms stopped at or below 300 epochs, against a validation curve that dips at 200 to 300 and then climbs past its early peak. No peak has been observed at any budget up to 10,000 epochs. Every per-arm delta on record measures early-training transients.
2. **Loss and metric point in opposite directions on expression, and only there.** Validation loss bottoms roughly 1,200 epochs before validation Pearson peaks. Every checkpoint the strand saved by loss was the wrong model. The 010 fitness and interaction task does not do this, so it is a property of this target or its objective.
3. **The perturbation operator provably cannot express a pair term.** At one perturbed gene the cross-attention softmax runs over a one-element key set, so the model reduces to $g(h_i + c_b)$ and nothing depends on which reporter is predicted given which gene was deleted. 16,200 of 32,760 attention parameters are dead. Capacity is not the binding constraint: a rank-32 reconstruction of the target reaches 0.727.
4. **Morphology has never been trained on its own data.** The Ohya screen holds 4,718 deletions; every morphology run used the 1,440 that also carry expression, giving $n_{\text{train}} = 1,161$. Its ceiling, measured here for the first time, is **0.611** over 278 features, with 201 of them individually above 0.5. For a second-label experiment, fitness shares 4,220 strains with morphology against expression’s 1,440.
5. **The free amino-acid profile predicts betaxanthin; the precursor does not.** Over the 4,432 deletions the two screens share, the 19-dimensional pool predicts betaxanthin at out-of-fold $r = \mathbf{0.298}$, while tyrosine alone, the chromophore precursor, predicts it at **0.064** and correlates marginally at -0.076 . Roughly three quarters of the profile’s power survives regressing out single-mutant fitness. On the 724 deletions with no Costanzo fitness record, which carry twice the betaxanthin spread, the profile predicts at 0.455.
6. **The metabolome auxiliary head currently costs betaxanthin performance.** Paired within grid cell, joint minus control is -0.0265 ± 0.0159 over the five comparable cells where the control learned the task, with 2 of 5 positive. The $+0.0203$ that motivated the replication reproduces, at $+0.0192$, in the one cell it came from. So the coupling exists in the data and the current head does not use it.

7. **Betaxanthin is the strongest case, and its strength is coverage.** yeast-GEM reaches 19% of the screen and misses 73% of its high producers. On genes a flux-based method has no features for, CGT's top 25 is enriched 4.4× over base rate. Where both methods can be scored they are tied, they find largely different genes (9 shared of 29 each), and their union reaches 49 of 100.
8. **Configuration choice moves results more than method choice.** Betaxanthin grid cells span Spearman 0.013 to 0.158 against a between-method gap of 0.0014 on AUC. The amino-acid strand's best run is 0.211 and its median is 0.0498.

1.3 The decision ✗

- **Figure 6 is ready to be argued** on coverage rather than on accuracy, and the amino-acid material belongs in it as a coupling measurement, not as a joint-training result.
- **Figure 3 is not ready, and its placeholder numbers are far from the measured ones.** A per-gene Pearson of 0.24 against a ceiling of 0.775 is not a headline. The cheapest path to a better one is not another architecture ladder: it is training morphology on its full dataset, finding where the expression curve actually turns, and checking whether the graph prior is the right prior at all.
- **Three cheap experiments are unrun and each would change what to do next:** the graph-prior probe, morphology at full scale, and one expression run trained until the validation curve turns.

2 Expression ✗

2.1 The task and its ceiling ✗

The **expression strand** predicts, for a strain carrying one gene deletion, the $\log_2$ ratio of every measured gene's transcript abundance against wild type. Targets come from the Kemmeren 2014 knockout microarray compendium (doi:10.1016/j.cell.2014.02.054) and the Sameith 2015 double-deletion compendium (doi:10.1038/sdata.2015.15), fused in the **fig3_core** build. Of the 1,556 records in that build, 1,484 are single deletions and 72 are Sameith doubles, so 95.4% of the training signal carries exactly one perturbed gene. SRC: experiments/019-simb-multimodal/results/fig3_overlap_census.json

The **reproducibility ceiling** is the highest correlation any predictor could reach against a target that carries measurement noise. It is estimated here from the 82 deletions measured independently in both compendia, as the mean over reporter genes of $\sqrt{\text{clip}(\hat{\rho}_g, 0, 1)}$ where $\hat{\rho}_g$ is that gene's test-retest correlation across the paired strains. The estimate is **0.7746**. SRC: experiments/019-simb-multimodal/results/expression_ceiling_replicate.json

The best validation score observed on this task is **pearson_per_feature = 0.2407**, from run **hx8pxdic** in project **torchcell_019_expr_v9**, whose peak sits at epoch 9,188 of the 9,997 it reached. The best in **torchcell_019_expr_v8** is 0.2293, run **b50f93ju**, peaking at epoch 4,007 of 4,105. Both are $\approx 30\%$ of the ceiling, and both peak in the last tenth of their run. SRC: experiments/019-simb-multimodal/results/round_leaderboards.csv

The metric is the per-reporter correlation across strains, averaged over reporters, which is the quantity that goes to zero when a model predicts each gene's mean; the companion per-strain metric stays near 0.4 under exactly that collapse and is not the objective. The scoring rule is **roll_max**, the maximum of a centered five-point rolling mean of the validation curve. It is an upward-biased order statistic whose bias grows with the number of epochs run, so it is comparable only between runs of similar length, which is why the epoch count travels with every number quoted from the leaderboard.

The manuscript's Results section currently states $r = 0.543$ for expression. That string sits inside a paragraph marked **[FILLER - R3.]** and is a placeholder, not a measurement.

2.2 The finding of the round: nothing was trained to saturation ✗

Sixty-seven runs across waves 1 to 4b were scored, and not one of them ran past 300 epochs. Against a validation curve that dips at epochs 200 to 300 and then climbs past its early peak, every one of those runs was stopped inside the dip. SRC: experiments/019-simb-multimodal/results/decoder_arms_torchcell_019_expr_v8.csv

Four long runs were later allowed to continue. All four died at 12.60 to 12.61 hours of wall clock at 1,620 to 1,625 epochs, which is a scheduler limit rather than convergence, and their validation peaks sit at 93 to 99% of the way through the run. Eval-mode training Pearson reached 0.652 to 0.725 and was still climbing when they were killed.

wave	runs	epoch range
wave 1	10	70–140
wave 2	5	76–140
wave 3	36	18–276
wave 4a	8	77–81
wave 4b	8	300
all	67	18–300, none above 300

Table 2: Every scored arm in the expression round stopped at or below 300 epochs.

Longer runs have since been allowed. The picture did not change: the best v8 run peaks at epoch 4,007 of 4,105 and the best v9 run at epoch 9,188 of 9,997. Raising the epoch budget from 1,600 to 10,000 moved the score from 0.204 to 0.241 and moved the peak with it, so the curve is still rising where it was cut off. **No peak has ever been observed on this task**, at any budget tried, and the saturation epoch remains unknown.

Consequence. An arm comparison at 300 epochs is not a small effect measured noisily. Both arms were stopped before either had the chance to separate, so the difference between them reflects early-training transients. The null sink arm reads $+0.0024 \pm 0.0090$ over four seed pairs and post-perturbation mixing reads -0.0023 ± 0.0076 , and neither number is evidence about its mechanism. More seeds do not repair this: replicates buy precision on the wrong estimand.

2.3 Loss and metric move in opposite directions ✗

Validation **expression/loss** reaches its minimum long before validation Pearson peaks, then rises for the remaining $\approx 1,200$ epochs while Pearson keeps climbing.

run	val loss min	val Pearson max
N_sink_mm s1	0.2640 @ ep 53	0.0640 @ ep 257
N_sink_mm s42	0.2653 @ ep 217	0.0824 @ ep 296
R_ref s1	0.2639 @ ep 42	0.0632 @ ep 293
R_ref s42	0.2662 @ ep 202	0.0701 @ ep 297

Table 3: Minimum validation loss precedes maximum validation Pearson by 40 to 250 epochs in wave 4b, and by $\approx 1,200$ epochs in the long runs. The 010 fitness and interaction task does not do this.

Two things follow. Every checkpoint the expression strand ever saved by validation loss was the early-dip model, roughly 1,300 epochs before the good one, which is why best-by-metric checkpointing was added. And the divergence itself is information about the objective: a quantile loss on a $\log_2$ ratio is minimized by a well-calibrated, narrow predictive distribution, while Pearson rewards getting the ordering right at whatever scale. The prediction standard-deviation ratio moving from 0.001 to 0.52 over the same window is consistent with the model widening its predictions long after the loss says it should stop.

What this does not license. The mechanism sentence above is a reading of the curves, not a measurement. Nothing has tested whether a different loss recovers the Pearson gain earlier. That is one of the options in section 8.

2.4 The perturbation operator has no pair term ✗

At $|S_b| = 1$ the cross-attention that carries the perturbation runs its softmax over a one-element key set, so the attention weight is identically 1. Re-drawing W_Q and W_K at standard deviation 10 changes the output by exactly 0.0, and 16,200 of 32,760 attention parameters are dead. SRC: experiments/019-simb-multimodal/results/perturbation_selector_degeneracy.json

The model therefore reduces to $g(h_i + c_b)$: gene identity i and strain identity b meet once, by addition, and no term depends on the pair (p, i) of deleted gene and reporter gene. The rank of the pair term each candidate mechanism can express is what wave 6 was built to ladder.

Capacity is not what binds. A rank- r reconstruction of the target matrix reaches 0.7265 at $r = 32$ and 0.7799 at $r = 64$, against a best model score of 0.241, so a null on this axis is a statement about the mechanism. SRC: experiments/019-simb-multimodal/results/lowrank_output_ceiling.json

mechanism	form	pair rank
V_ref additive	$g(h_i + c_b)$	0
V_sink gated attention	one bounded scalar per head	9
V_basis {16, 32, 64}	$\sum_j b_{ij}(h_i) a_j(z_S)$	r
V_film , V_hadamard	diagonal multiplicative	$d = 90$

Table 4: Pair-rank ladder. The reference arm cannot express any dependence on which reporter is being predicted given which gene was deleted.

2.5 The largest signal in the strand is imputation, and it is orthogonal ✗

A ridge oracle that observes m randomly chosen genes of a held-out strain and predicts the rest reaches far past anything the model does.

m observed	within-Kemmeren	within-Sameith	Kem $\rightarrow$ Sam	Sam $\rightarrow$ Kem
10	0.4562	0.3649	0.2335	0.2122
100	0.6693	0.6417	0.3815	0.3922
1000	0.7832	0.7779	0.4838	0.4803

Table 5: Masked-conditioning oracle, `pearson_per_feature` on validation strains. The within-study value at $m = 1000$ exceeds the 0.775 reproducibility ceiling, which the cross-study column resolves: same-array conditioning was predicting the target’s own measurement noise.

The gain is not a better genotype-to-expression map. Removing a genotype predictor, a k -nearest-neighbor model on `prot_T5_all` with $k = 25$, leaves 97.5 to 100.6 % of the conditioning gain intact, so the channel the objective optimizes is switched off at $m = 0$, which is where the strand is scored. SRC: `experiments/019-simb-multimodal/results/conditioning_gain_after_genotype.json`

What makes the oracle worth keeping is the structure it exploits. After removing each gene’s mean, the residual gene-gene correlation pattern replicates on held-out strains at split-half $r = 0.8687$ against a permutation null of 8.45×10^{-5} , and it has an effective rank of 32.78, with 59.1 % of variance inside the top 32 components. A per-gene independent readout emits 6,127 marginals and discards all of it. SRC: `experiments/019-simb-multimodal/results/residual_covariance_diagnostic.json`

2.6 Why the byte count is misleading ✗

The expression compendium holds more scalar labels than the trigenic interaction data the model does well on, which is the intuition that predicted this task would be easy. The label count is not the axis that matters.

dataset	records	per record	scalars	perturbation axis
Kuzmin 2018 trigenic interaction	91,111	1	91,111	triples; a gene recurs with many partners
Kemmeren 2014 knockout expression	1,484	6,169	9,153,196	1,484 single deletions, each once
Ohya 2005 morphology	4,718	281	1,325,758	4,718 single deletions, each once

Table 6: Scalar labels are not the binding resource. Expression carries ≈ 9.2 million scalars against the trigenic set’s 91 thousand, but its perturbation axis has 1,484 points and no gene is deleted twice, so nothing in the data isolates the effect of a deletion from the effect of the one array it was measured on.

In the trigenic data a given gene appears in many records paired with different partners, so the supervision repeatedly constrains that gene’s contribution. In the expression data a gene appears as the perturbation exactly once. The 6,169 labels attached to it are 6,169 readouts of a single perturbation event, correlated with each other at the level the residual covariance above measures, so they are worth far less than 6,169 independent constraints on the deletion.

Established. The expression strand sits at 0.241 against a ceiling of 0.775. The round produced no valid comparison between mechanisms, because every arm was scored before saturation. It produced three things that hold: the additive operator provably cannot express a pair term; loss and metric diverge on this target and not on 010; and the one large available signal is imputation from other genes of the same strain, measured to be orthogonal to the genotype-only task.

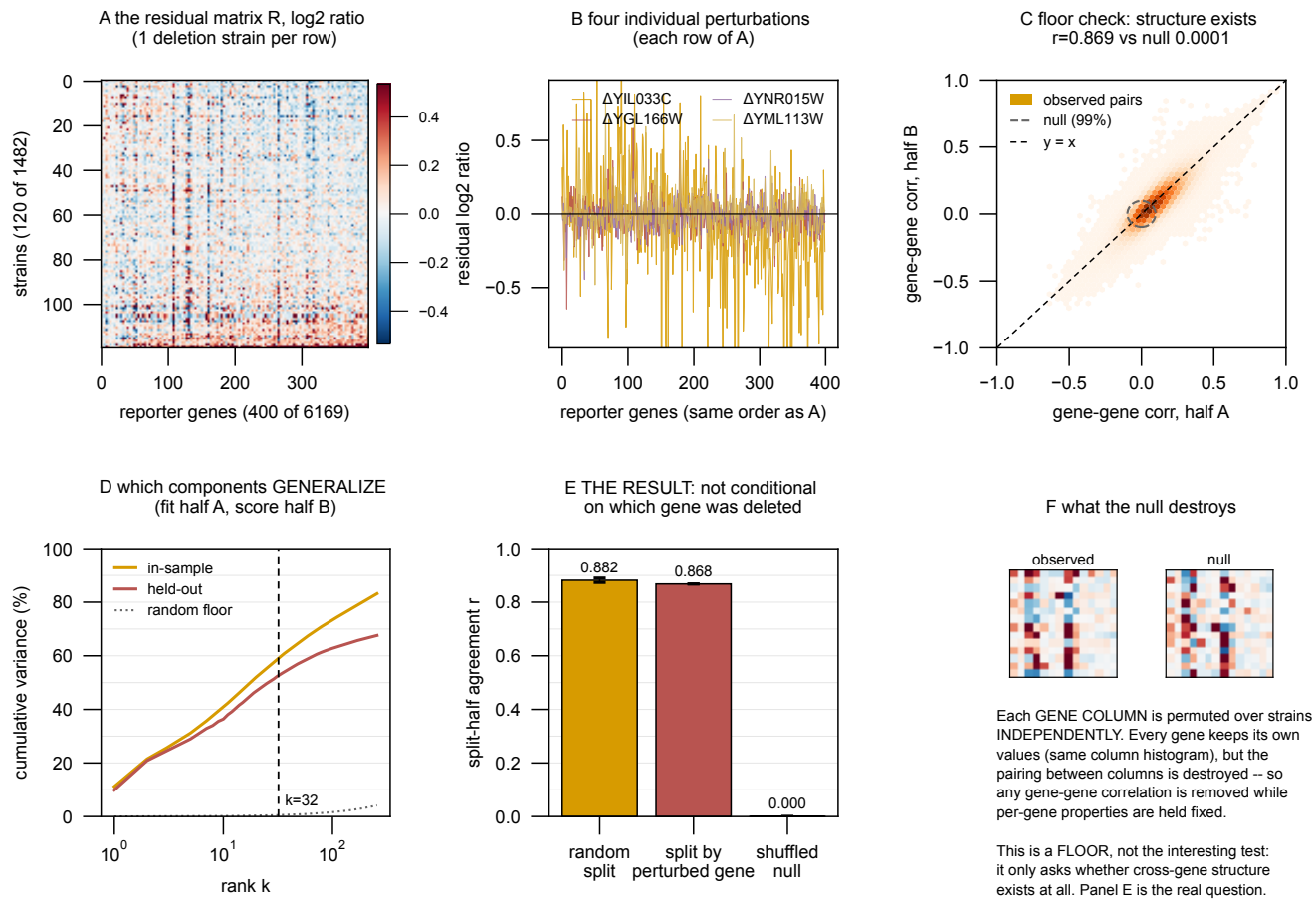


Figure 1: Residual gene-gene structure in the expression compendium after removing each gene's mean. The pattern replicates on held-out strains and lives in about 33 effective components, which is what a per-gene independent readout discards.

Cannot say. Whether any of the pair-rank mechanisms help. Whether the graph prior is the right prior for this task. Whether a different objective closes the loss-metric gap.

3 Morphology ✗

3.1 The target ✗

The **morphology strand** predicts the CalMorph phenotype vector of a single-deletion strain. CalMorph’s nominal vocabulary is 501 parameters, which resolve into 281 base parameters and 220 coefficient-of-variation statistics; the served target is the 281 base parameters, of which the model scores 278 after three near-degenerate features are dropped. Source is Ohya 2005 (doi:10.1073/pnas.0509436102), 4,718 deletion mutants plus 122 independent *his3* wild-type replicates.

Raw CalMorph values span about eight orders of magnitude, from 6.7×10^{-5} to 1.49×10^4 in absolute mean, so an unnormalized mean-squared-error loss is dominated by whole-cell size and actin-region size and read ≈ 4.7 million. The training path now applies a per-feature z -score fit on the training split only, which brings the loss to order 1 and makes “predict each feature’s mean” score zero rather than win. Ohya’s own normalization is a per-parameter Box-Cox fit on wild-type data followed by standardization, and it discards 247 of 501 parameters on a Shapiro-Wilk normality test at $p \geq 0.5$. That verdict is a normality test over base and CV parameters together, not a variance floor on the 278 modeled here, so it was not imported.

3.2 The ceiling was never measured until now, and it is high ✗

Ohya gives one biological sample per mutant but 122 wild-type replicates, so per-feature reliability is estimable as $1 - \sigma_{\text{WT}}^2 / \sigma_{\text{mutants}}^2$ and the achievable Pearson is its square root.

quantity	value
mutants	4,718
wild-type replicates	122
modeled features	278
mean reliability	0.444
mean ceiling over modeled features	0.611
median ceiling	0.647
features with ceiling > 0.5	201 of 278 (72 %)
best observed (vsceij2v, peak epoch 27 of 65)	0.0824
fraction of ceiling realized	13.5 %

Table 7: Morphology has the second-highest ceiling of any strand here and the lowest fraction of it realized. Nearly three quarters of the modeled features are individually measurable to $r > 0.5$.

3.3 The strand was never really run, on two counts ✗

Twenty-three runs exist in `torchcell_019_morph_v5`, spanning 11 to 121 epochs, one of them a collapsed constant predictor. The best peaks at epoch 27 of 65. The joint expression-plus-morphology project is thinner still: 32 runs, none past 87 epochs, best `val/mean/pearson_per_feature` of 0.0505.

Set against section 2, where the same architecture on a related target was still improving at epoch 9,188, a 65-epoch morphology run carries no information about whether morphology is learnable. The morphology result is not a null. It is an absence of measurement, and the two must not be recorded the same way.

And the training set was a quarter of what exists. Every morphology run in the project reports $n_{\text{train}} = 1,161$ and $n_{\text{val}} = 132$. The Ohya screen holds 4,718 deletions, and only 1,440 of them also carry expression. The morphology configs set `require_modalities:[expression_log2_ratio,calmorph]`, which pins every arm to that intersection.

SRC: `experiments/019-simb-multimodal/conf/delta_joint_expr_morph_000.yaml`

That restriction is correct *for the auxiliary-task question*, since comparing a joint model against a morphology-only model on different instance sets would confound the auxiliary head with a data-quantity difference. It is wrong for the question “how well can morphology be predicted”, and the same configs were used for both. **Morphology has never been trained on its own full dataset.**

3.4 Which second label should morphology be paired with ✗

The overlap census answers this directly, and the answer is not expression.

genotypes carrying	count
morphology (Ohya, total)	4,718
morphology and fitness	4,220
morphology and expression	1,440
morphology, fitness and expression	1,326

Table 8: Fitness shares nearly three times as many strains with morphology as expression does, and covers 89 % of the morphology screen against expression’s 31 %.

Pairing morphology with fitness keeps 4,220 of the 4,718 strains, so the joint arm and the morphology-only arm can be run on a large common instance set without the restriction above costing three quarters of the data. Pairing it with expression caps both arms at 1,440.

This does not retire the expression-helps-morphology question. That question is what `delta_joint_expr_morph_000.yaml` was built to answer, its control is right, and its three conditions (`expr`, `morph`, `expr_morph`) are the correct decomposition. It has simply never been run past 87 epochs. What the census changes is the order: the fitness pairing is where the statistical power is, and it can be run at full scale first.

What the earlier “0.05 to 0.14 rise” was. A flatten-Pearson computed over the $[B, 281]$ matrix before per-feature normalization, which is dominated by between-feature scale: large features are large in both prediction and target. After z -scoring, the same smoke run reports ≈ 0 on that metric. The honest metric is the per-feature correlation across strains, averaged over features, which is what table 7 scores.

3.5 A scalar target is a reasonable next step, and this is which scalar ✗

Ranking single features on reliability alone selects ones that are precisely measured and nearly constant across mutants. The rank key below is ceiling multiplied by robust coefficient of variation, IQR over absolute median across the 4,718 mutants, so a feature has to be both measurable and moved by deletions.

feature	CalMorph label	ceiling	robust CV	score
A113	actin_n_ratio	0.873	2.369	2.068
C124_C	Medium_bud_ratio	0.698	0.886	0.618
A110_A1B	Actin_f_ratio	0.637	0.939	0.598
C125_A1B	Large_bud_ratio	0.784	0.630	0.494
A105	actin_a_ratio	0.655	0.617	0.404
D212	nuclear_B_ratio_to_nuclear_AA1BC_cells	0.604	0.642	0.387

Table 9: Top of the scalar-target shortlist. The list is dominated by bounded ratios (actin localization class, bud-size class, nuclear-stage class), which is what a deletion actually moves. Size features are the opposite trade: C115_A whole-cell axis ratio has a ceiling of 0.972 and almost no spread across mutants.

On “predict colony size first”. Colony size is fitness, and fitness is already the strongest strand in the project, trained on the Costanzo and Kuzmin single-mutant fitness data. Using it as the morphology warm-up would test the pipeline rather than the phenotype. Within CalMorph the equivalent of a colony-size scalar is a cell-size feature, and those are exactly the features deletions barely move. **Recommendation: run the scalar warm-up on A113 or C124_C**, both of which are reliable and genuinely variable, and treat the run as a convergence-budget calibration for the vector target rather than as a result.

Established. Morphology has a mean ceiling of 0.611 over 278 features with 201 of them individually above 0.5, and the best score achieved is 0.0824, at 13.5 % of it. That score comes from a 1,161-strain training set drawn from a 4,718-strain screen. Target normalization is settled and the earlier apparent rise was a scale artifact.

Cannot say. Anything about whether morphology is learnable from genotype, because no morphology run has passed 121 epochs or seen more than a quarter of the data. Anything about whether expression and morphology share structure, because the joint project’s longest run is 87 epochs.

4 Betaxanthin production ✗

4.1 The task, its ceiling, and the two names that get confused ✗

Cachera is a genome-wide betaxanthin deletion screen (doi:10.1093/nar/gkad656): the four-gene betalain cassette *CYP76AD1*, *DOD*, *ARO4*^{K229L} and *ARO7*^{G141S} was transferred by CRI-SPA into each strain of the yeast knockout collection and per-colony fluorescence read as a production proxy. It supplies 4,735 single deletions and is the ground truth here. **Flux Cone Learning** (FCL) is Merzbacher et al.’s method, which learns from flux samples drawn through a genome-scale metabolic model; **RandomForestClassifier_Resampled** is its best variant. The comparison is CGT against FCL, both scored on the Cachera screen.

Betaxanthin is the only phenotype in this document with a high ceiling. Its records carry a per-strain standard error over a median of 15 colonies, giving a reliability of 0.836 and a Pearson ceiling of **0.914**. SRC: experiments/019-simb-multimodal/results/pigment_noise_ceiling.json

The best validation Pearson recorded on this target is 0.430 in study **betaxanthin_002**, at 47% of that ceiling, with the top five configurations in close agreement rather than one lucky trial. The cell carried forward for the head-to-head below is **s09_L6_maskon_1r0.0001_yj**, selected by validation at 0.3639 and never by its score on the held-out comparison genes.

4.2 The published accuracy comparison is empty for both methods ✗

FCL reports gene-level accuracy 0.700 against a majority-class rate of 0.673, achieved by calling 607 of 640 genes medium, which is 94.8%. Our absolute binning sits at 0.681 doing the same thing. Forced to the true class marginal, FCL drops to 0.556 and CGT to 0.549.

The structural point. Every model that finds more high producers has worse overall accuracy, because the only way to call more highs is to stop calling everything medium. Accuracy selects against the capability the task is about, so it should not be the primary metric in any writeup, including ours.

4.3 Both methods have real signal, and it lives in the tail ✗

Of the top k genes each method nominates, the fraction that are truly high producers, against a base rate of 0.156:

model	$k = 10$	$k = 25$	$k = 50$
FCL RF	8 ($5.1\times$, $p = 10^{-5}$)	19 ($4.9\times$, $p = 8 \times 10^{-12}$)	20 ($2.6\times$, $p = 10^{-5}$)
CGT	9 ($5.8\times$, $p = 4 \times 10^{-7}$)	18 ($4.6\times$, $p = 10^{-10}$)	24 ($3.1\times$, $p = 2 \times 10^{-8}$)

Table 10: Precision at k for high producers. Nine of CGT’s top ten are genuine high producers. Both curves approach the base rate by $k = 150$.

Median percentile rank by true class is flat for both methods, at 54/49/56 for FCL and 48/51/52 for CGT across low, medium and high. Read with the table above, that says the signal is real and confined to the tail, not that there is none.

4.4 The two methods find different genes ✗

Rank-matched into the same 100 high slots, FCL recovers 29 of the 100 true high producers and CGT recovers 29, and they share 9. The union reaches 49. Their rank correlation is -0.128 , which at $n = 639$ is $z = -3.2$, where two models correlating with the truth at $+0.105$ and $+0.039$ would through the truth alone correlate with each other at about $+0.004$.

No mechanism is claimed. for the sign. What is claimed is the operational consequence: recall is near-additive because the two methods are finding largely different genes, which is an argument for running both, and which is invisible in every aggregate statistic. The 51 that neither finds is the honest ceiling on this comparison.

4.5 The real difference is coverage ✗

yeast-GEM contains 1,161 genes, of which 915 appear in the Cachera screen. That is 19.4% of its 4,721 deletions. A flux-based method has features for those and for nothing else.

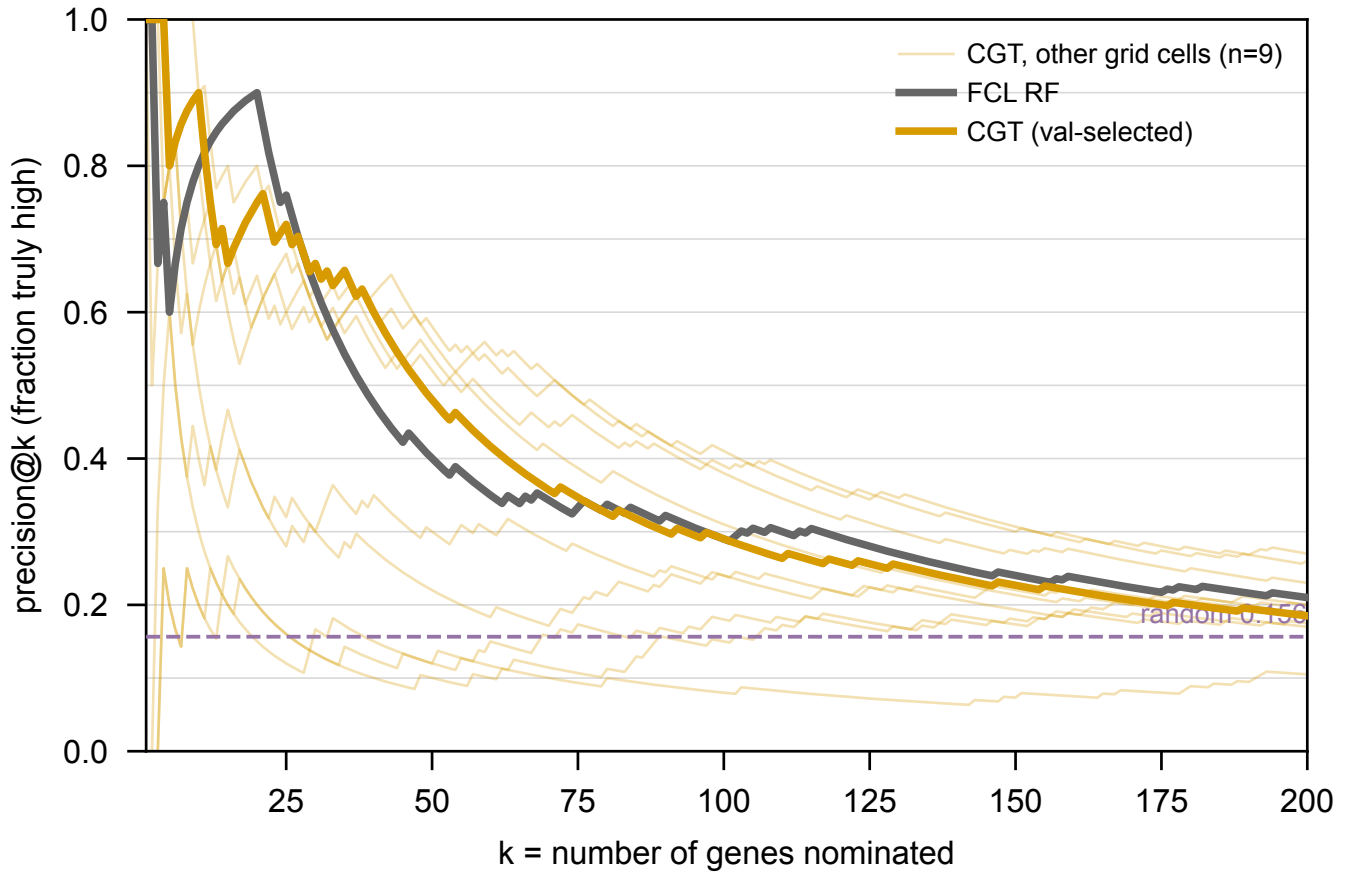


Figure 2: Precision at k for high producers, both methods, against the base rate of 0.156. Nine of CGT's top ten are genuine high producers, and both curves reach the base rate by $k = 150$. The nine unselected CGT grid cells span 0.10 to 0.65 at $k = 50$, so cell choice matters more than the gap between methods.

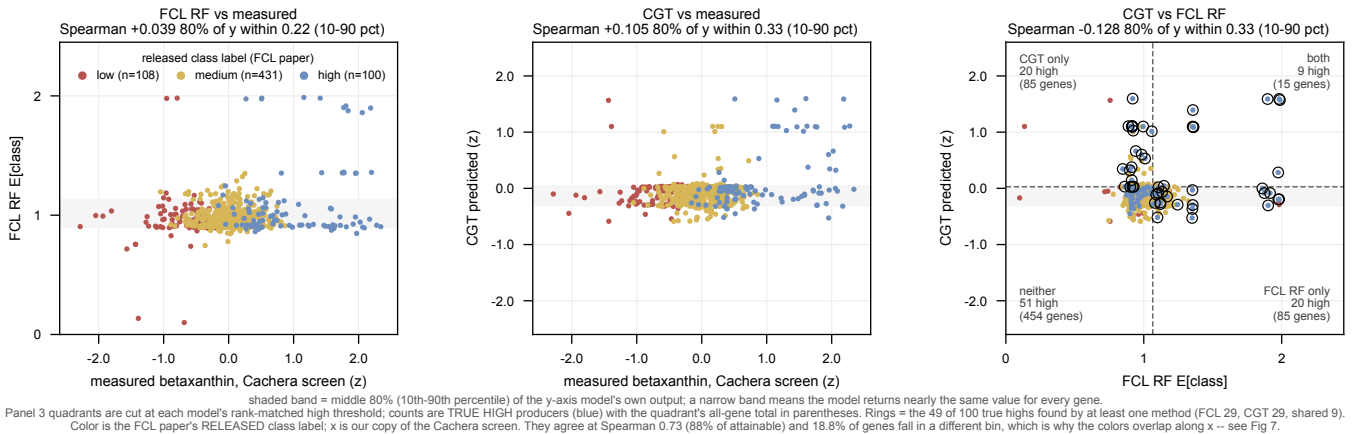


Figure 3: Per-gene spread over the 639 shared genes. Both models emit a nearly constant value: 80 % of FCL's genes fall inside an expected-class band of 0.22 on a 0 to 2 scale, 80 % of CGT's inside 0.33 z -units. Panel c shows the disagreement concretely, with each method recovering 29 of the 100 true high producers and sharing 9.

population	in yeast-GEM	outside
every deletion in the screen	915 (19 %)	3,806 (81 %)
top 100 measured producers	32 (32 %)	68 (68 %)
all high producers	60 (27 %)	163 (73 %)

Table 11: Metabolic genes are enriched among high producers, at 26.9 % of the highs against 19.4 % of the screen, a 1.4× enrichment that is exactly what the biology predicts. It is not enough to matter: restricting to metabolism buys that concentration and costs three quarters of the hits.

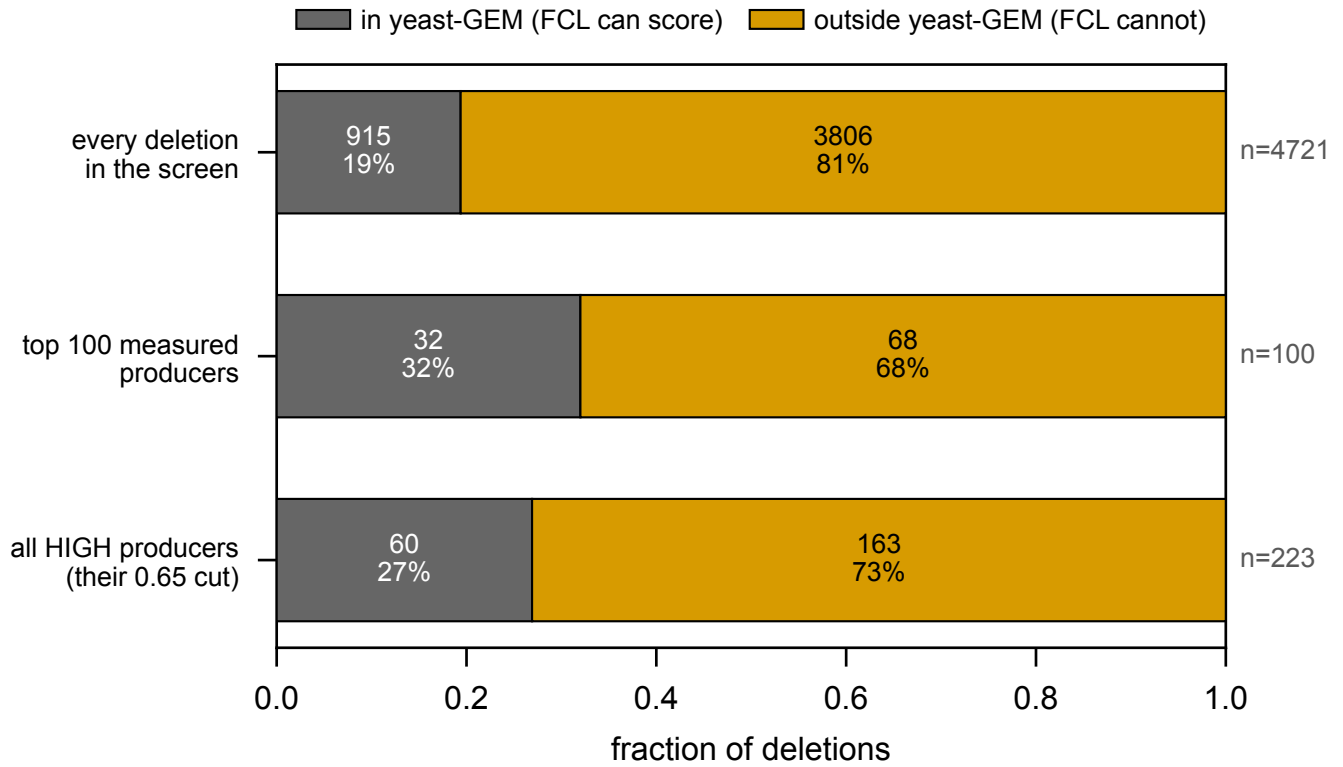


Figure 4: Measured data only, no model and no prediction: how little of the betaxanthin screen a flux-based method can reach.

On held-out test genes split by yeast-GEM membership, with identical derived labels on both sides, CGT's top 25 recovers 17 of 48 true highs inside the model ($9.3\times$ over base rate, $p = 2 \times 10^{-15}$) and 9 of 21 outside it ($4.4\times$, $p = 2 \times 10^{-5}$).

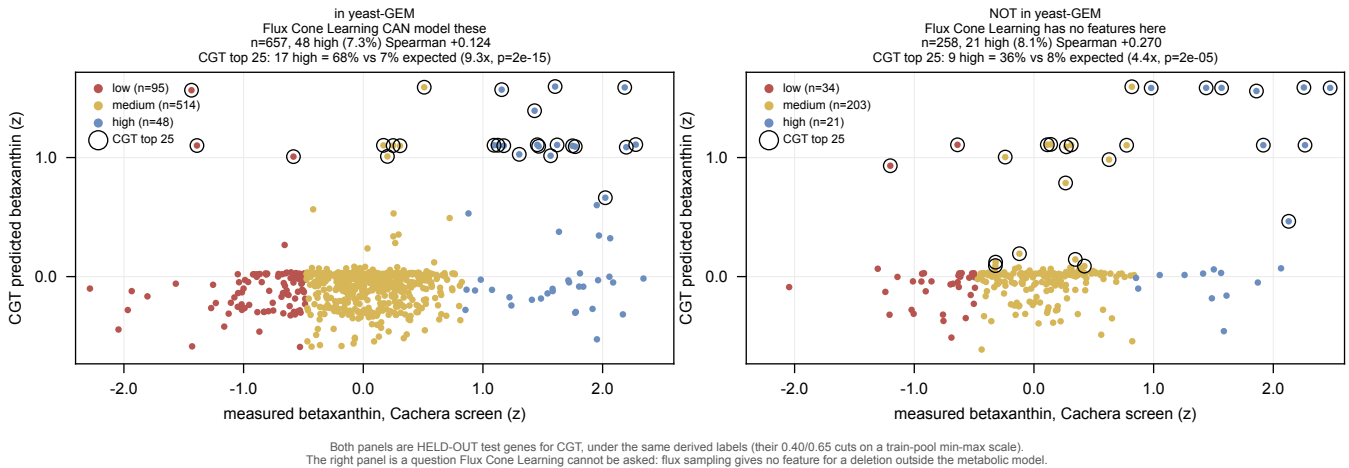


Figure 5: CGT predicts high betaxanthin producers among deletions a flux-based method cannot represent. Held-out test genes split by yeast-GEM membership, identical derived labels on both sides, circled points are CGT's top 25. Flux Cone Learning has no prediction for the right panel.

FCL's prediction for the outside set is not poor, it is undefined: flux sampling yields no feature for a deletion the metabolic model does not contain. The FCL test set is 639 of 640 inside yeast-GEM; of our other test genes only 21 of 294 are.

Two qualifications that must travel with this claim. The rank correlation reads higher outside the model than inside ($+0.270$ against $+0.124$), but the two are measured on different, non-overlapping gene sets and a Fisher r -to- z gives $z = 2.07$, $p = 0.039$. Read it as at least as good outside as inside. And labels for the non-GEM genes are derived by us, since the FCL paper never labeled them, so the comparison to make is panel against panel.

4.6 Where the comparison is soft ✗

- **The selection is asymmetric, and it favors FCL.** Ours is the validation-selected cell. Theirs is the best of four models, justified by a test-set number, and their validation folds appear to be drawn from the same 640 genes as their test list. Where the two land level, we are level with an opponent picked with more information than we allowed ourselves.
- **Validation Pearson does not track test AUC.** Four of ten CGT cells beat FCL’s AUC of 0.570 and the best reaches 0.695, while the validation-selected cell reads 0.557. The runner-up on validation, 0.3528 against 0.3639, has an AUC 0.14 higher. Neither “CGT’s AUC is 0.695” nor “CGT’s AUC is 0.557” is safe; the metric does not separate the two methods at this resolution.
- **Their labels are a headroom bound.** Their binning rule min-max scales production and cuts at 0.40 and 0.65, so applied to our larger copy of the same screen it gives 107/476/56 against their 109/431/100. 18.8 % of genes sit in a different bin than our values would give, so a model predicting our measurement perfectly is marked wrong on about a fifth of genes.
- **Regression against classification.** They tried regression and abandoned it, in their words “challenging with the limited number of knockouts at the high and low ends.” Our ceiling of 0.914 makes regression available to us. The comparison is therefore between a regressor and a three-class classifier, which is why every metric above is either rank-based or computed after binning our predictions with thresholds fit on the training split only.
- **The build is stale** with respect to the gene-name resolver (issue #195), which the split builder avoids by working from the raw screen, and which the training data still inherits until a rebuild.

Established. On this screen CGT and FCL are tied where both can be measured, both have tail-confined signal at roughly 5× enrichment in the top 25, and they find different genes. CGT predicts the 81 % of the screen a flux-based method cannot represent, and finds high producers there at 4.4× over base rate.

Cannot say. That CGT beats FCL on any single aggregate. That the rank anti-correlation means anything mechanistically. That the absolute top-100 membership is a deployable call, since the rank-matched comparison hands both sides the true class marginal.

5 Amino-acid pools, and whether they help production ✗

5.1 The target ✗

Muller 2016 (doi:10.1016/j.cell.2016.09.007) measured the intracellular concentration of 19 amino acids in 4,678 single-deletion strains. There is one replicate per strain and no released standard error, so the dataset admits **no within-dataset reproducibility ceiling**. Every correlation involving it is attenuated by an unmeasured amount, and that qualification travels with every number below.

The best validation `pearson_per_feature` is **0.211**, run `24vlgope` in `torchcell_022_muller19_v4`, peaking at epoch 402 of 999. The median across the 31 runs is 0.0498, so the leaderboard is a long tail under one good configuration rather than a plateau. Earlier rounds read ≤ 0.025 , then 0.171, so the trajectory is upward across rounds.

5.2 These runs converge, which the expression runs never do ✗

Twenty-four runs in the preceding round were scored for their convergence shape. The typical run peaks at epoch 48 to 134, continues for another 41 epochs with validation slope turning negative while training slope stays positive, and is classified converged then overfitting. None of them hit the epoch cap. SRC: `experiments/022-muller-metabolome/results/training_length_analysis.json`

This is the opposite failure to section 2, where nothing peaked at any budget up to 10,000 epochs. The two strands run the same architecture on the same encoder, so whatever makes expression climb for thousands of epochs is a property of the expression target or its loss, not of the model.

5.3 The precursor hypothesis is wrong, and the profile hypothesis is right ✗

Betalains derive from L-tyrosine (tyrosine to L-DOPA to betalamic acid), and a betaxanthin is betalamic acid condensed with an amino acid. The mechanistic prediction is that a deletion’s free amino-acid pool carries information about how

much betaxanthin that same deletion makes. Both screens are keyed by systematic ORF and share 4,432 deletions, so the prediction is directly testable with no model in the loop.

predictor of betaxanthin	out-of-fold Pearson r	n
tyrosine alone (the precursor)	0.064	4,432
19 amino acids jointly	0.298	4,432
<i>on the 3,708 deletions with a Costanzo fitness record</i>		
single-mutant fitness alone	0.151	3,708
19 amino acids	0.176	3,708
19 amino acids and fitness	0.200	3,708
19 amino acids, fitness regressed out of both	0.133	3,708
<i>on the 724 with no Costanzo record</i>		
19 amino acids	0.455	724

Table 12: Cross-validated ridge, five folds, three shuffles; the spread across shuffles is ≤ 0.022 everywhere and ≤ 0.004 on the large sets. Amino acids enter as log concentration, z -scored.

Tyrosine is not the axis. Its marginal correlation with betaxanthin is $r = -0.076$ ($p = 3.9 \times 10^{-7}$), which is negative, and it ranks sixth of nineteen by absolute correlation. The strongest is methionine at $+0.140$, followed by aspartate $+0.127$ and arginine -0.118 . No amino acid exceeds $|r| = 0.15$. Correcting for the betaxanthin side’s known reliability of 0.836 raises these by a factor of 1.094, which does not change the reading; the Muller side’s correction is unmeasurable and would raise them further by an unknown amount.

The profile is the axis. The 19-dimensional pool predicts betaxanthin at out-of-fold $r = 0.298$, roughly $4.7\times$ the precursor alone. For scale, the CGT’s own genotype-to-betaxanthin validation Pearson is 0.372 to 0.430, so a linear map from 19 measured metabolite concentrations recovers most of what the model recovers from genotype.

It is not all growth. Single-mutant fitness alone predicts betaxanthin at 0.151, so a shared growth component is real and substantial. Regressing fitness out of both sides leaves the amino-acid profile at 0.133, so roughly three quarters of its predictive power on that gene set survives the control. Combining the two reaches 0.200, above either alone.

And most of the signal is where fitness is unavailable. The 724 deletions with no Costanzo KanMX record have a betaxanthin standard deviation of 0.914 against 0.464 for the rest, and on them the amino-acid profile predicts at 0.455. This is the population that carries the extremes, and it is the population a fitness-based control cannot reach.

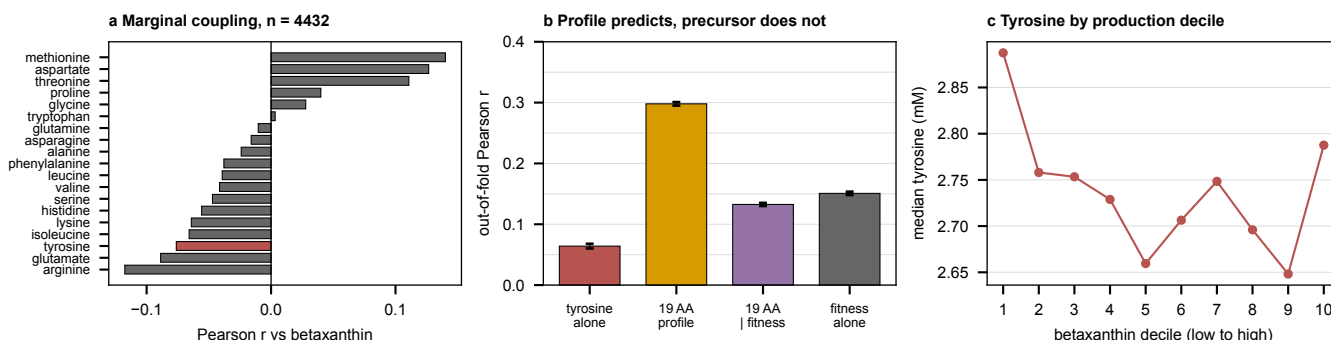


Figure 6: (a) Marginal correlation of each amino acid with betaxanthin over the 4,432 shared deletions; tyrosine highlighted. These reproduce `experiments/019-simb-multimodal/results/pigment_noise_ceiling.json` exactly, which the generating script asserts rather than assumes. (b) Cross-validated ridge fits from table 12. (c) Median tyrosine by betaxanthin decile. The relationship is non-monotone: tyrosine is highest in the lowest-production decile, falls through the middle, and rises again at the top, which is why a linear correlation reads near zero. The axis spans 2.65 to 2.89 mM, so the swing is about 9% of the median.

A mechanism, labeled as unverified. The Cachera background carries $ARO4^{K229L}$ and $ARO7^{G141S}$, feedback-resistant alleles that release the shikimate pathway from the control that shapes the pools Muller measured in the plain deletion collection. A plausible reading is that free tyrosine in an unengineered strain is not the quantity that limits flux in an engineered one. This has not been tested and no experiment here can test it.

5.4 Does the metabolome head actually help the betaxanthin head $\times$

The claim under test is narrow and was stated before the round: in Delta grid cell `s04_L6_maskon_1r0.0001`, the joint arm read $+0.0203$ over the control on `val/betaxanthin/pearson_per_feature` at a matched budget of 333 epochs.

That was one paired difference at one seed with no estimate of the noise floor on it. The same grid read -0.049 and -0.060 in the two $L = 2$ cells, both run to 1,000 epochs, so the sign flips with depth.

grid cell	control	joint	Δ	aux r	epochs (ctrl/joint)
s00_L2_maskon_lr0.0001	0.3927	0.3473	-0.0455	0.149	999/999
s02_L2_maskoff_lr0.0001	0.3570	0.3078	-0.0492	0.053	999/999
s04_L6_maskon_lr0.0001	0.3769	0.3960	+0.0192	0.113	999/999
s06_L6_maskoff_lr0.0001	0.4216	0.3609	-0.0608	0.061	999/999
s08_L4_maskon_lr0.0001 [†]	0.4105	0.4199	+0.0094	0.120	999/747
s10_L4_maskoff_lr0.0001	0.3705	0.3745	+0.0039	0.054	999/999
s01_L2_maskon_lr0.001*	0.0553	0.0399	-0.0155	0.029	999/999
s03_L2_maskoff_lr0.001 [†] *	0.0400	0.0474	+0.0074	0.013	490/999
s05_L6_maskon_lr0.001*	0.0674	0.0126	-0.0548	0.012	999/999
s07_L6_maskoff_lr0.001*	0.0699	0.0587	-0.0112	0.009	999/999
s09_L4_maskon_lr0.001*	0.0461	0.0529	+0.0068	0.009	999/999
s11_L4_maskoff_lr0.001*	0.0611	0.0478	-0.0133	0.010	999/999
mean, cells where the control learned (5 cells)		-0.0265 ± 0.0159		2 of 5 positive	
mean, all matched-exposure cells (10 cells)		-0.0220 ± 0.0090		3 of 10 positive	

Table 13: Betaxanthin score with and without a 19-amino-acid auxiliary head, paired within grid cell, on `val/betaxanthin/pearson_per_feature`. Δ is joint minus control. *aux* is the auxiliary head’s own score, carried because an auxiliary head at $r \approx 0$ that still moved the primary metric would mean the gain came from regularization rather than from shared metabolic signal, and those are different findings. [†] marks a cell whose two arms ran epoch counts differing by more than 50; since the score is a running maximum, such a difference is part effect and part unequal exposure, and those rows are excluded from both means. * marks a cell whose control arm scored below 0.10, meaning neither arm learned the task; every one of them is an $\text{lr} = 10^{-3}$ cell, and a difference between two failed runs measures nothing about the auxiliary head.

Scored uniformly across the whole grid, the effect is **negative**. Over the five matched-exposure cells whose control arm learned the task at all, the mean difference is -0.0265 ± 0.0159 (standard error, $n = 5$), with 2 of 5 positive. Over all ten matched-exposure cells including the collapsed ones it is -0.0220 ± 0.0090 , with 3 of 10 positive.

Three readings the table supports, in decreasing confidence.

1. **The original +0.0203 reproduces, and it is one cell.** Cell `s04_L6_maskon_lr0.0001` reads +0.0192 here under a different scoring rule and a different epoch budget, which is close agreement. It is also the only clearly positive live cell, against two large negatives at $L = 2$ and one at $L = 6$ with the graph mask off.
2. **Depth and the graph mask both matter more than the auxiliary head.** The two $L = 2$ cells read -0.0455 and -0.0492 ; the two $L = 6$ cells read +0.0192 with the mask on and -0.0608 with it off. Any single-cell claim about the metabolome head is smaller than the spread across the cells it could have been made in.
3. **The auxiliary head is weak wherever the primary is good.** Its own score runs 0.053 to 0.149 across the live cells, well under the 0.211 the amino-acid strand reaches when it is the only head. So the joint arm is paying capacity for a head that is itself underperforming, which is a different problem from the coupling being absent.

The measurement this still lacks. Every number in the table is $n = 1$ per arm per cell except `s08`, so the spread across cells mixes the effect with initialization noise, and no within-cell noise floor exists. That is exactly the gap the replication config was written to close.

The designed replication, `experiments/023-metabolome-betaxanthin-joint/conf/igb_bx_aa.yaml`, adds a third setting: an auxiliary head restricted to **tyrosine only**, against a common control, on a gene set pinned by `require_head_targets` so no arm can win by seeing more rows. **It has not been run**; no W&B project `torchcell_023_bx_aa_v1` exists. Given table 12, its tyrosine arm is now predicted to be the null one and its 19-amino-acid arm the live one, which is a sharper prediction than the config was written against.

Established. The free amino-acid profile of a deletion predicts that deletion’s betaxanthin production at $r = 0.298$ out of fold, tyrosine alone predicts it at 0.064, and about three quarters of the profile’s power survives regressing out single-mutant fitness. The amino-acid strand itself reaches 0.211 and converges rather than under-trains.

Scored across the whole 023 grid, adding the metabolome head *lowers* betaxanthin by -0.0265 ± 0.0159 over five comparable cells. The one positive cell is the one the original claim came from.

Cannot say. How much of the 0.298 is attenuated by Muller’s unmeasurable noise, so it is a floor rather than an estimate. Whether a shared encoder can exploit the coupling: the data says the coupling is there, the joint training says the current head does not use it, and at $n = 1$ per cell those two are not yet in contradiction.

The gap this exposes. A linear map on 19 measured metabolite concentrations reaches 0.298 on betaxanthin, and the auxiliary head that is supposed to inject the same information reaches 0.053 to 0.149 on those metabolites and then costs the primary task 0.027. The coupling is not the bottleneck. The route by which a second head’s representation reaches the first head’s readout is.

6 Beta-carotene production ✗

6.1 Why this target is different from the other pigment ✗

Ozaydin 2013 (doi:10.1016/j.ymben.2012.07.010) screened the deletion collection carrying a beta-carotene pathway and scored each colony’s color **by eye on an ordinal scale from -5 to $+5$** . The dataset holds 4,474 records, of which 4,344 have a single replicate and 130 have two or three. That is a different kind of measurement from the Cachera fluorescence readout, and it needs a different ceiling: a Pearson ceiling on a subjective ordinal is the wrong object, so rank agreement is used.

Two estimates exist, and they disagree by a factor of seven.

estimate	basis	n	Spearman
replicate max against min	130 replicated rows	130	0.544
independent re-screen	2ndRoundOfTransformations sheet	119	0.075
range-restriction corrected	Thorndike case 2, Pearson	119	0.105

Table 14: Reliability of the beta-carotene score. Neither estimate is clean. The max-against-min comparison is biased in both directions at once: the two are drawn from one replicate set, which inflates agreement, and they are the widest possible split of that set, which deflates it. The re-screen is a genuine test-retest but was run on selected top hits, so its 1st-screen scores sit almost entirely in $3 \dots 5$ of a $-5 \dots +5$ scale.

The reported ceiling is 0.544, from the replicate comparison. Read the 0.075 as what a re-screen of selected winners looks like under range restriction, not as evidence that the score is worthless.

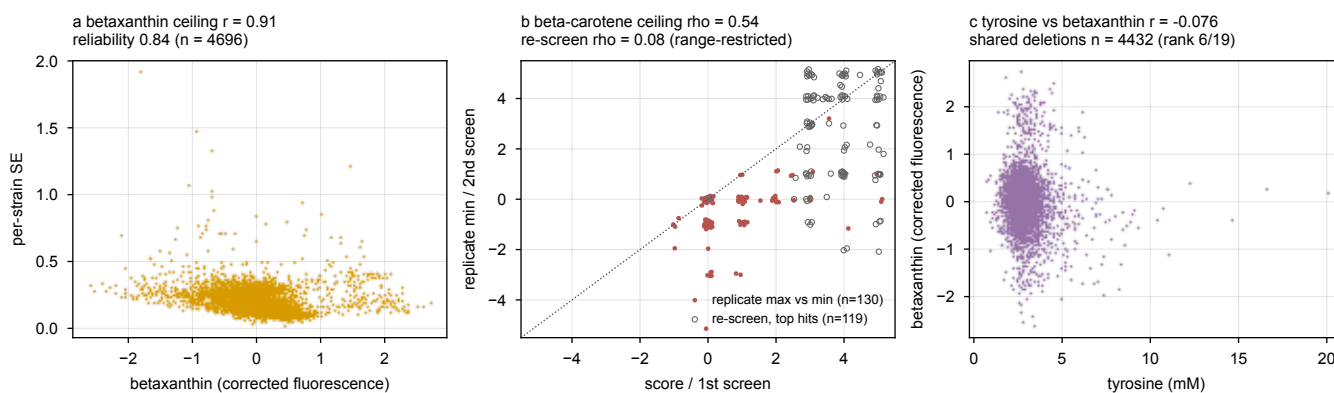


Figure 7: Reliability of both pigment targets. The betaxanthin panel uses the per-record standard error over a median of 15 colonies; the beta-carotene panels use rank agreement, because the target is a hand-scored ordinal.

6.2 What was measured ✗

Thirty-nine runs in `torchcell_021_beta_carotene_v4`, spanning 134 to 999 epochs. The best validation `spearman_per_feature` is **0.132**, run `o0aadmot`, peaking at epoch 799 of 999. Median across live runs is 0.072. Against the 0.544 ceiling that is 24% realized.

Established. Beta-carotene is the weakest pigment target and the reason is visible in the measurement rather than in the model: a hand-scored ordinal with one replicate per strain has a reliability that no architecture recovers. The strand does carry signal, at 0.132 against a chance level of zero, and its best run peaks late (epoch 799 of 999), so it shares the expression strand’s under-training rather than the metabolite strands’ early overfitting.

Cannot say. Which of the two ceiling estimates is right, and therefore whether 0.132 is a quarter of achievable or most of it. That gap is resolvable only with a replicate design the source data does not contain.

Recommendation. Keep beta-carotene as a second production target for the coverage argument in section 4, where the claim is about which genes a flux model can represent and does not depend on the absolute score. Do not spend a sweep on lifting its number.

7 What the strands have in common ✗

7.1 One table, read the same way ✗

strand	ceiling	best	of ceiling	epochs (peak/last)	runs
Expression (Kemmeren, Sameith)	0.775	0.229	30%	4007/4105	163
Expression, masked-label objective	0.775	0.241	31%	9188/9997	16
Morphology (CalMorph, Ohya)	0.611	0.082	13%	27/65	23 (1 flat)
Expression and morphology, joint	–	0.051	–	28/70	32
Betaxanthin (Cachera)	0.914	0.372	41%	254/999	44
Betaxanthin with metabolome head	0.914	0.430	47%	176/496	46
Amino-acid pools (Mulleder)	–	0.211	–	402/999	31
Beta-carotene (Ozaydin)	0.544	0.132	24%	799/999	39

Table 15: Every strand, scored the same way. *best* is `roll_max`, the maximum of a centered five-point rolling mean of the validation curve, an upward-biased order statistic whose bias grows with epochs run, which is why the epoch column travels with it. *epochs (peak/last)* gives where the best run peaked and where it stopped: a peak in the last tenth means the run was cut off while still improving. Ceilings are measured by three different estimators because the three measurement types require them; the amino-acid cell is empty because Mulleder has one replicate per strain, and no ceiling is estimable. *flat* counts runs whose validation metric never left zero.

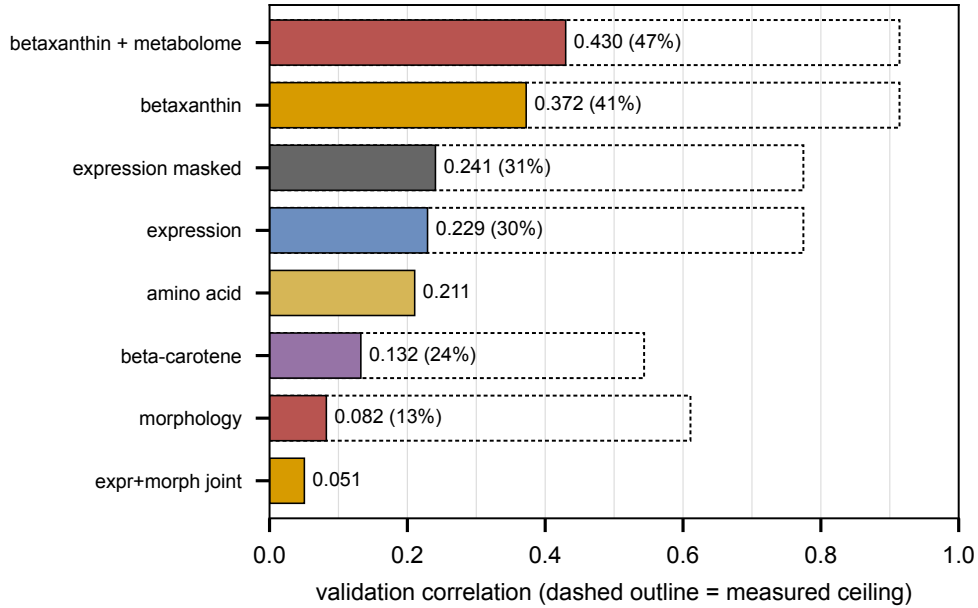


Figure 8: Achieved score against measured ceiling, best run per strand. The dashed outline is the ceiling and the gap is the headroom; amino acid and the expression-morphology joint arm carry no outline because no single ceiling applies to them. A bare correlation is not comparable across these rows, which is the reason the panel exists.

Three things are visible only when the strands sit together. The ceilings differ by more than a factor of two, so an unqualified r is uninterpretable across rows. The epoch budgets differ by more than a factor of a hundred, so the score column is partly a record of how long each strand was allowed to run. And the two strands closest to their ceilings, betaxanthin and amino acid, are the two with a scalar or low-dimensional target.

7.2 Two opposite convergence failures, and neither was diagnosed from a curve ✗

- **Expression never peaks.** At every budget tried, up to 10,000 epochs, the validation Pearson peaks in the last tenth of the run. Raising the budget from 1,600 to 10,000 raised the score from 0.204 to 0.241 and moved the peak with it.

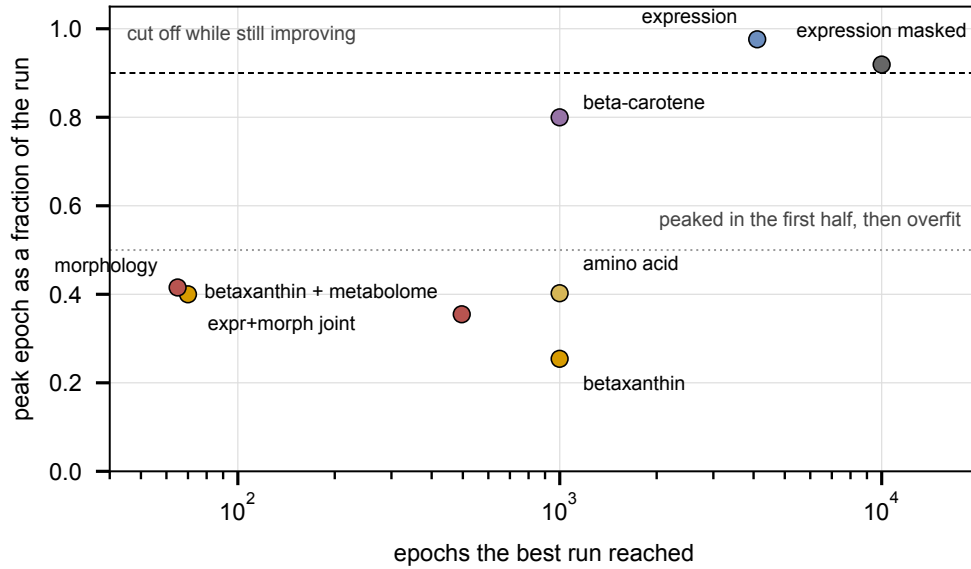


Figure 9: Where the best run’s peak sits inside its own budget. Points near the top were still improving when the run stopped, so their scores are lower bounds set by compute. Points in the lower band peaked early and spent the rest of the budget overfitting.

- **The metabolite strands peak early and then overfit.** Amino-acid runs peak at epochs 48 to 134 with validation slope turning negative while training slope stays positive. The best betaxanthin run peaks at epoch 254 of 999.
- **Morphology was never run long enough to tell which it does.** Longest run, 121 epochs.

Same encoder, same optimizer family, opposite behavior. The difference is in the target: a 6,127-dimensional vector of $\log_2$ ratios against a scalar or a 19-vector. Any single epoch budget applied across strands is wrong for at least two of them, which is what produced the truncated expression waves and would produce overfit metabolite runs if the budget were set from expression.

7.3 Loss and metric point in different directions on expression alone ✗

Validation loss bottoms 40 to 250 epochs before validation Pearson peaks in wave 4b, and $\approx 1,200$ epochs before it in the long runs, while the prediction standard-deviation ratio climbs from 0.001 to 0.52. The 010 fitness and interaction task does not do this: there both losses fall monotonically and the validation-loss minimum is at the last epoch. The consequence already cost the project every checkpoint the expression strand saved before best-by-metric checkpointing landed.

What this suggests, unverified. A quantile loss on a $\log_2$ ratio can be minimized by a narrow, well-calibrated predictive distribution that gets the ordering wrong, and Pearson is invariant to the global rescale that the loss is not. Reporting **nmse** alongside, defined so that **nmse** = 1 is exactly “predict each gene’s mean”, separates right-ordering from right-magnitude. That instrument exists; the experiment that would test whether a different loss closes the gap does not.

7.4 Configuration choice moves results more than method choice ✗

On betaxanthin the ten grid cells span Spearman 0.013 to 0.158 and AUC 0.406 to 0.695, against a between-method gap of 0.0014 on AUC. Four low points are the $\text{lr} = 10^{-3}$ cells, which collapsed to a constant predictor. On the amino-acid strand the best run is 0.211 and the median is 0.0498.

Any single-number claim about a method is dominated by which cell is quoted, which is why the cell must be fixed by validation in advance, and why the leaderboard records the median alongside the maximum. It also means an arm difference smaller than the cell spread is not measurable without pinning the cell and replicating seeds inside it.

7.5 Cross-modality overlap is real and small ✗

Before spending compute on joint proteome and expression training, the linear overlap between the two was measured on the 1,350 strains they share. Per-gene correlation across strains has median $r = 0.08$, with 1.7% of genes above 0.3 and 8.9% negative. A ridge map recovers held-out R^2 of 0.035 in one direction and 0.024 in the other, above a trivial baseline of -0.004 but far from redundancy. SRC: `experiments/019-simb-multimodal/results/proteome_expression_eda.json`

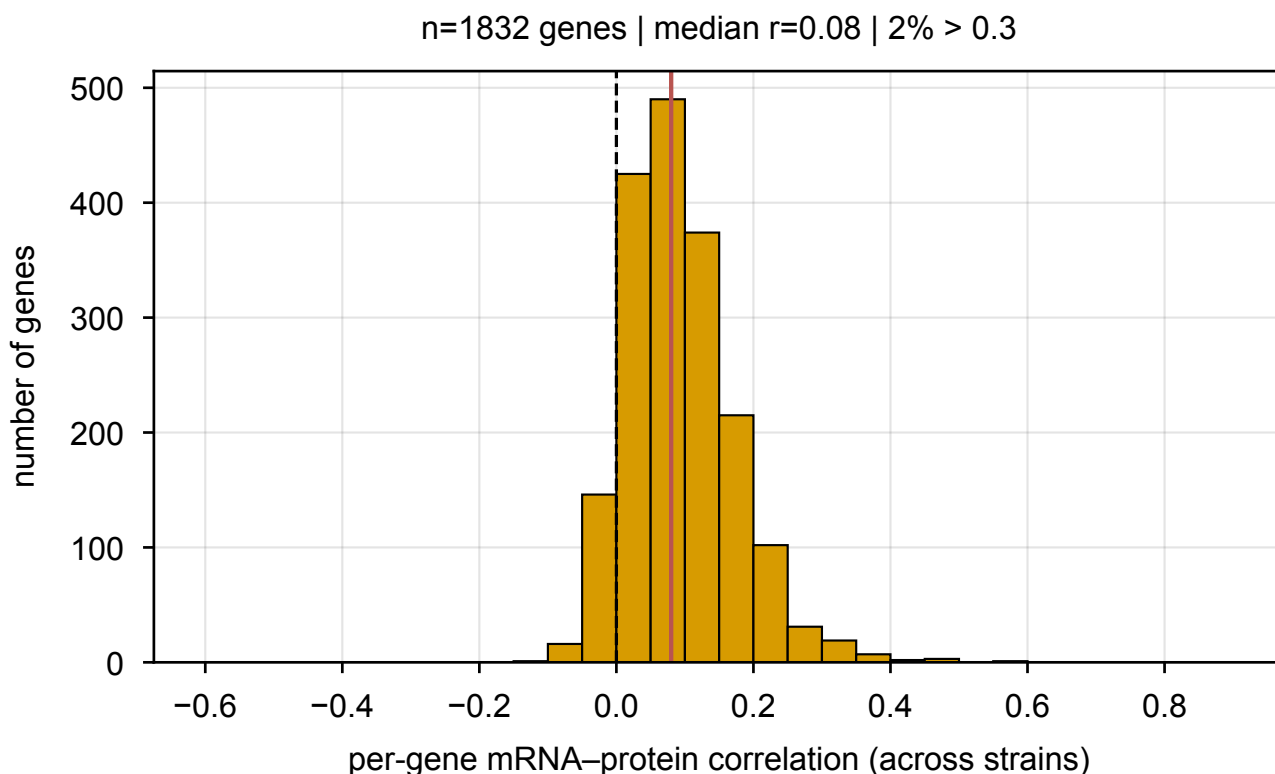


Figure 10: Per-gene correlation across strains between the Messner proteome and the Kemmeren expression compendium, over the 1,832 genes and 1,350 strains they share. A narrow lump barely off zero. The few high- r genes are dominated by the strain in which that gene is itself deleted, which drives both its transcript and its protein toward zero.

The two modalities were measured in different media, synthetic minimal for the proteome and synthetic complete for the expression, which attenuates every number above. The overlap is therefore certified as non-redundant and not certified as cleanly complementary. That distinction is what bounds how hard a “the proteome helps expression” claim can be pushed until a same-media pair exists.

7.6 The graph prior has never been checked ✗

The nine interaction graphs enter the model as a hard additive attention mask, so composing L layers makes the influence of gene X on gene Y approximately the L -step random-walk reachability of Y from X on that graph. That is a falsifiable assumption about the data: *deleting X perturbs the genes near X on graph k* . If it is false, pulling attention toward the graph pushes the model toward the wrong prior and no setting of the regularization weight rescues it.

The probe that would test it is specified and unbuilt. It costs minutes of CPU: for each deleted gene, the across-reporter rank correlation between graph proximity and response magnitude, reported per graph as an AUC against degree-preserving rewiring and a matched random graph, where 0.5 means the graph says nothing. SRC: `notes/experiments.019-simb-multimodal.graph-prior-probe.md`

Nine graphs consume nine attention heads. If three carry signal, the other six are constrained toward noise and are worse than free heads. This is the cheapest unrun experiment in the project.

7.7 The perturbation axis is the binding resource ✗

Table 6 makes the point for expression: 9.2 million scalar labels across 1,484 perturbations, each gene deleted exactly once. Morphology is the same shape, 1.3 million labels across 4,718 perturbations, each once. The trigenic interaction

data is the opposite shape, 91 thousand labels across 91 thousand records in which a given gene recurs with many partners.

The strands that work in this project are the ones where the supervision constrains a gene repeatedly. **Hypothesis, untested:** the reason multi-task training is attractive here is not that the labels are informative about each other, but that a second phenotype gives a second, differently-shaped constraint on the same perturbation. That predicts the gain should be larger where the two phenotypes share strains and smaller where they merely share genes, which is a testable statement and has not been tested.

8 What to run next ✗

Ordered by cost against what each would settle. The first four are cheap enough that not running them is the expensive choice.

8.1 Tier 0, cheap and decisive ✗

1. **The graph-prior probe** (section 7). Minutes of CPU, no GPU, no training. It asks whether deleting gene X actually perturbs X 's neighbors on each of the nine graphs, which is the assumption the entire graph channel rests on and which has never been checked. Nine attention heads are currently spent enforcing it. If three graphs carry signal, regularize those three and free the other six.
2. **Morphology at full scale.** Drop `require_modalities` for the morphology-only arm and train on all 4,718 Ohya deletions instead of the 1,440 that also carry expression. This is a config change, and it is the only way the 0.0824 becomes a number about morphology rather than about a quarter of morphology. Run it long enough to see a peak, which section 7 says cannot be assumed from the expression budget.
3. **The scalar morphology warm-up**, on A113 (ceiling 0.873, robust CV 2.37) or C124_C (Medium bud ratio, 0.698 and 0.886). Cheap, and it calibrates the epoch budget for the 278-feature target. Not colony size: colony size is fitness, which is already the project's strongest strand, and the CalMorph size features are the ones deletions barely move.
4. **Dump the missing convergence curves.** The expression strand has never observed a peak at any budget up to 10,000 epochs. Before another arm comparison is run on that target, one configuration should be trained until the validation curve turns, and the epoch at which it turns should set every subsequent budget. Any arm scored before that number is known is not a measurement.

8.2 Tier 1, the two questions worth GPU ✗

A. Does a second phenotype help a first one, tested where the power is. The 023 result says the metabolome head costs betaxanthin -0.0265 ± 0.0159 over five cells at $n = 1$ each, so the question is open and currently answered in the wrong direction. Two changes make the next attempt readable.

- **Pin the cell and replicate seeds inside it**, which is what `igb_bx_aa.yaml` already specifies and what none of the existing data does. Without a within-cell noise floor, a ± 0.016 difference cannot be distinguished from initialization.
- **Pair morphology with fitness rather than expression** for the first full-scale test. Fitness shares 4,220 strains with morphology against expression's 1,440, so both arms can run at three times the instance count. Then run the expression pairing on the 1,440 as the mechanism test it was designed to be.

Substituting tyrosine for the 19 amino acids, as `igb_bx_aa.yaml` proposes, is now predicted to be the *null* arm rather than the live one, since tyrosine alone predicts betaxanthin at 0.064 against the profile's 0.298. That prediction is worth keeping in the design as a negative control, not as the headline contrast.

B. Is the expression objective the problem. Loss and metric point in opposite directions on expression and on nothing else in the project. Three arms, at a budget set by Tier 0 item 4:

- the current quantile loss, as reference;
- a loss computed on per-gene z -scored targets, so that “predict each gene's mean” scores zero rather than winning, which is already implemented as `standardize_per_feature_target`;

- a correlation-based objective, which optimizes the quantity being scored.

The reason to run this before another mechanism ladder: a mechanism arm evaluated under an objective that is optimizing something else cannot separate “the mechanism does not help” from “the objective does not reward it”.

8.3 Tier 2, the structural options ✗

Constraint-based coupling, and what it would actually buy. Adding a constraint-based metabolic layer, in the physics-informed sense of penalizing predictions that violate a stoichiometric or flux constraint, is attractive for expression on the argument that metabolism constrains it. Two facts bear on the cost side. First, yeast-GEM covers 19.4 % of the deletions in the betaxanthin screen and 27 % of its high producers (table 11), so a metabolic constraint is silent on most of the genome. Second, the strand where a metabolic constraint would be most directly testable, betaxanthin, is already the strand closest to its ceiling. **Recommendation: treat the constraint layer as a Fig-6 mechanism, not as an expression rescue**, and gate it on whether the coupling it assumes is visible in the data the way table 12 made the amino-acid coupling visible.

Reifying the gene node into mRNA and protein. Not yet. The measured case for it is thin: per-gene mRNA-to-protein correlation across strains has a median of 0.08, a ridge map between the two recovers 2 to 3.5 % of held-out variance, and the two datasets were measured in different media, so the decoupling that would justify separate nodes is partly a condition artifact. Splitting the node multiplies parameters on an axis the data does not yet resolve. The case changes when a same-media proteome and expression pair exists, or when perturbations become fine-grained enough that the central-dogma collapse is what breaks.

Masked-label imputation as its own deliverable. The masked-conditioning oracle reaches cross-study 0.4838 at $m = 1000$ against the model’s 0.241, and 97.5 to 100.6 % of that gain survives removing a genotype predictor. It is a real capability and a defensible product surface, and it is not a genotype-to-expression result. Deciding whether to ship it as a separate thing is a decision, not an experiment.

8.4 What to do about Figure 3 and Figure 6 ✗

Figure 3. The manuscript currently claims $r = 0.543$ for expression and 0.619 for morphology inside a paragraph marked [FILLER – R3.]. The measured values are 0.241 and 0.082. Neither placeholder should survive into a draft anyone reads. The honest version of the panel is the ceiling-relative one: expression at 31 % of a measured 0.775, morphology at 13 % of a measured 0.611 on a quarter of its data, with the fitness panel carrying the strong result. **A 0.20 per-gene Pearson is not a Figure 3 headline**, and the two Tier 0 morphology items are the cheapest route to finding out whether a better one exists.

Figure 6. The betaxanthin case is the strongest thing in this document and it does not depend on beating anyone on accuracy. The argument is coverage: yeast-GEM reaches 19 % of the screen and misses 73 % of its high producers, and CGT finds high producers in the part a flux method cannot represent at 4.4× over base rate. Panel order that carries it: setup diagram, betaxanthin performance against its 0.914 ceiling, the coverage gap, the capability gap, and the FCL comparison last, framed as a tie on the genes both can score. The amino-acid panel is currently a negative and should be shown as the coupling measurement (fig. 6) rather than as a joint-training result.